

We begin by observing that the SMM uses the Helmholtz free energy of Eq. (13) to define the entropy of Eq. (18) and uses the energy condition of Eq. (17) to determine the configuration temperature T_0 . It then defines the statistical weight of a configuration as

$$w_n \Delta \varepsilon_0 = \exp(S_n(T_0)) , \quad (25)$$

where $\Delta \varepsilon_0$ is a small interval in energy about the total value ε_0 . In most cases, the differences between the statistical weight of the FBM, Eq.(22), and that of the SMM, Eq. (25) are irrelevant, as the variations in specific heats and temperatures among configurations are small compared to the exponential variation with respect to the entropy[25].

In fact, although we entered into great detail to calculate the density of states of an FBM fragment configuration in terms of the quantities used in the SMM, the association between the two models was already established once we identified the statistical weight of the FBM as the density of states. The microcanonical SMM defines the statistical weight of a fragment configuration in terms of the microcanonical entropy S_n , which, in turn, can be defined in terms of the density of states ω_n as

$$S_n = \ln(\omega_n \Delta \varepsilon_0) . \quad (26)$$

Thus, the FBM and the SMM use the same physical quantities in much the same way. They still have their differences, however, which we discuss next.

A minor distinction between the FBM and the SMM, as they are commonly used, is their treatment of the normalization volume V_n . Both models use a normalization volume larger than the volume of the decaying nucleus V_0 , with the difference expressed in terms of a multiplicative factor χ . As mentioned before, calculations with the FBM model often use a normalization volume twice that of the volume of the decaying nucleus[8],

$$V_{n,FBM} = (1 + \chi) V_0 \quad \text{with} \quad \chi = 1 . \quad (27)$$

SMM calculations normally use a factor of χ between 2 and 5, but exclude the volume of the fragments[18, 25], so that

$$V_{n,SMM} = \left((1 + \chi) V_0 - \sum_{j=1}^n V_j \right) = \chi V_0 \quad \text{with} \quad \chi = 2 - 5 . \quad (28)$$

As defined here, the volumes in both models are independent of the fragment configuration and are obviously the same when $\chi = 2$ is used in the SMM. In Ref. 19, a multiplicity-dependent volume was used in order to assure that the fragments were formed outside